

AI-Assisted Pet Allergy Symptom Tracking System

Sam Cashion | 700619652 | scc96520@ucmo.edu

The purpose of this database application is to track and analyze allergy-related symptoms in pets over time. This project is motivated by my personal experience managing chronic allergies in my own dog, Finn. Finn experiences recurring allergy symptoms such as severe itching and skin irritation that fluctuate daily. Without a structured way to record this information, it is difficult to identify patterns, evaluate symptom severity over time, or clearly summarize his allergy history.

The database is needed to provide a consistent and organized method for recording allergy symptoms and transforming daily observations into meaningful information. The system allows users to create pet profiles and log daily allergy symptoms, including the symptom type, severity level, date, and optional notes. This data enables the system to support queries that analyze symptom trends and summarize allergy activity over a selected number of days.

The primary users of the system are pet owners managing pets with chronic or recurring allergies. These users need to understand what symptoms their pet has experienced, how often those symptoms occur, and whether symptom severity is improving or worsening over time. The system addresses the problem of fragmented or memory-based tracking by storing symptom data in a structured relational database that supports historical analysis.

Input data available to the database includes pet information and daily symptom entries provided by the user. Each symptom entry consists of a date, a symptom category, a severity rating, and optional descriptive notes. The database stores this information in a time-stamped format to allow for aggregation, comparison, and trend analysis.

To improve accessibility, the system integrates a Large Language Model (LLM) that enables users to interact with the database using natural language. Users can ask questions such as “What allergy symptoms has Finn had in the last 30 days?” or “Has his itching worsened this week?” The LLM translates these questions into structured database queries and returns clear, user-friendly summaries of the stored data.